

RAJAT KHANDURI

Software Engineer

 [wthrajat.com](https://github.com/wthrajat)
 India / Remote

 [wthrajat](https://www.linkedin.com/in/wthrajat)
 wth.rajat@gmail.com
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EXPERIENCE

Software Engineer

Amboss Technologies  [ambosstech](https://ambosstech.com)

Nov 2024 – Present

United States / Remote

- Own identity and paid access across every Amboss product: designed and built the OAuth 2.0 single sign-on platform with TOTP two-factor, plus the L402/LSAT pay-per-use layer that monetizes API endpoints without subscriptions.
- Build product features end-to-end through the stack - GraphQL and REST APIs in NestJS paired with React/Next.js frontends - working directly with product and business teams.
- Contribute across the Bitcoin and Lightning platform: Taproot Assets surfaces, Lightning liquidity products, and LND integrations behind Amboss market data and payment tools.
- Deliver on AWS and Kubernetes with CI/CD pipelines, and build the internal tooling the engineering team runs on.

Software Engineer / Maintainer

Openswap  [citadel-foss/openswap](https://citadel-foss.org/openswap)

Nov 2023 – Oct 2024

United States / Remote

- Maintained Openswap (formerly CoinSwap), a privacy-focused Bitcoin atomic-swap protocol, as a core Rust library maintainer for a year - owning wallet storage, protocol encoding, public APIs, documentation, and integration tests.
- Made wallets materially cheaper to run: reworked storage onto binary CBOR encoding and cut wallet storage by 28.8%.
- Improved the contributor and CI feedback loop by redesigning protocol message encoding and decoding, reducing integration-test latency by 21.7%.
- Drove a maintainability push through the Rust core - removed legacy paths, modularized the architecture, and overhauled wallet configuration.
- Owned the project's public Rust API documentation.

Selected Fellow / Open Source SWE

Summer of Bitcoin  [citadel-foss/openswap](https://citadel-foss.org/openswap)

May 2024 – Sep 2024

United States / Remote

- Rebuilt the Openswap wallet from scratch on BDK's chain interfaces and data structures, replacing the legacy wallet implementation.
- Reimplemented WalletStore, Wallet, and the public API surface around modular storage designed for long-term maintainability.
- Extended protocol key management with Fidelity and SwapCoin keychain variants, and integrated RPC synchronization into the new wallet.

Selected Fellow / Open Source SWE

Google Summer of Code  [xwiki-contrib/application-interactive-link-visualizer](https://xwiki-contrib.org/application-interactive-link-visualizer)

May 2023 – Sep 2023

France / Remote

- Built an interactive graph explorer that visualizes how pages link across large XWiki instances, shipped as an installable extension.
- Integrated XWiki platform APIs to extract backlinks, titles, URLs, and document references dynamically, layered with search, facets, filters, and theme-aware graph controls.
- Documented the extension publicly and produced the project blog post and video walkthrough.

Freelance Software Engineer

Freelance  [wthrajat](https://wthrajat.com)

Dec 2021 – Apr 2023

India / Remote

- Ran client engagements end-to-end through Upwork, Fiverr, and Freelancer - shipping products like GreatWall and iWriting alongside hackathon builds.
- Built and maintained small web products, APIs, frontend surfaces, and automation workflows.
- Merged 2000+ pull requests across open-source and private work.

TECH SKILLS

Languages	TypeScript, Rust, Go, Python, JavaScript, C/C++, Solidity, Shell
Backend	Node.js, NestJS, gRPC, REST, GraphQL, PostgreSQL, Redis, OAuth 2.0, JWT
Frontend	React, Next.js, Astro.js, D3.js, Graphology, SigmaJS
DevOps	AWS, Docker, Kubernetes, CI/CD, Datadog, GitHub Actions
Blockchain	Bitcoin, Lightning Network, Taproot Assets, Payments, Wallets, Protocol testing
AI	MCP, RAG, LLM APIs, Claude Code, Codex
Styling	Tailwind CSS, Design systems

ACHIEVEMENTS

Summer of Bitcoin	selected from 5000+ applicants.
Linux Foundation Mentorship	selected from 10,000+ applicants.
Google Summer of Code	selected fellow from the top 2% of 43,765 applicants; shipped a 350-hour project.
Major League Hacking	selected for the software engineering track.
CodeChef Long Challenge II	Global Rank 1.
Open-source and private work	2000+ total pull requests merged.
Technical writing	maintains technical docs and writes on systems, Go, and web development.

PROJECTS

unixdm

 [wthrajaat/unixdm](#)

→ C++17 terminal download manager with segmented downloads, resumable state, batch queues, and an interactive TUI.

C++17, CMake, POSIX, FTXUI, Shell

- Built a C++17 terminal download manager with segmented downloads, batch queues, scheduling, and resume files.
- Implemented worker rebalancing, retries, preallocation, and an FTXUI interface with live download feedback.

Coreloop

 [wthrajaat/coreloop](#)

→ Self-hosted learning and news platform with a Next.js control surface, durable Go services, and Telegram delivery.

Go, Next.js / TypeScript, Turso / libSQL, Telegram Bot API, QStash, Groq / Gemini

- Designed Go services and a Turso-backed durable queue for scheduling, leases, retries, provider health, and delivery progress.
- Built a Next.js control surface and Telegram delivery path with source-backed content, Groq/Gemini routing, and deterministic fallbacks.

autocommit-rs

 [wthrajaat/autocommit-rs](#)

→ Published Rust CLI for generating conventional commit messages from staged Git changes.

Rust, Git, OpenAI, Gemini, Cargo

- Built and published a Rust CLI for conventional commit generation from staged Git changes.
- Added interactive setup, OpenAI/Gemini support, configurable message style, signing, and hook controls.

Interactive Link Visualizer

 [xwiki-contrib/application-interactive-link-visualizer](#)

→ XWiki extension for searchable page relationship graphs.

TypeScript, XWiki, Solr, Graphology, Sigma.js

- Integrated backlinks, titles, URLs, and document references through XWiki APIs.
- Added search, facets, filters, theme-aware graph UI, documentation, and demos.

Seamcarving

 [wthrajaat/seamcarving](#)

→ C++ content-aware image resizing with OpenCV and CMake.

C++, OpenCV, CMake, GLEW

- Implemented seam identification and removal with dynamic programming.
- Built a C++ binary around OpenCV, GLEW, and CMake.

AA Torque for Honda Elevate

 [wthrajaat/android-auto-torque](#)

→ Extended AA Torque for Honda Elevate with Android Auto dashboard customization, gauges, conversions, and docs.

Kotlin, Java, Android Auto, Torque Pro, OBD2

- Customized AA Torque for Honda Elevate with Android Auto themes, gauges, and conversion strings.
- Documented Honda Elevate setup and Torque Pro/OBD2 workflow while keeping fork attribution clear.

EDUCATION

Bachelor of Engineering, [Panjab University](#)

Year: 2025, Grade: 7.25/10

High School, [CBSE](#)

Year: 2020, Grade: 9.56/10